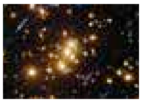
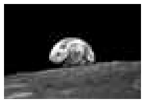
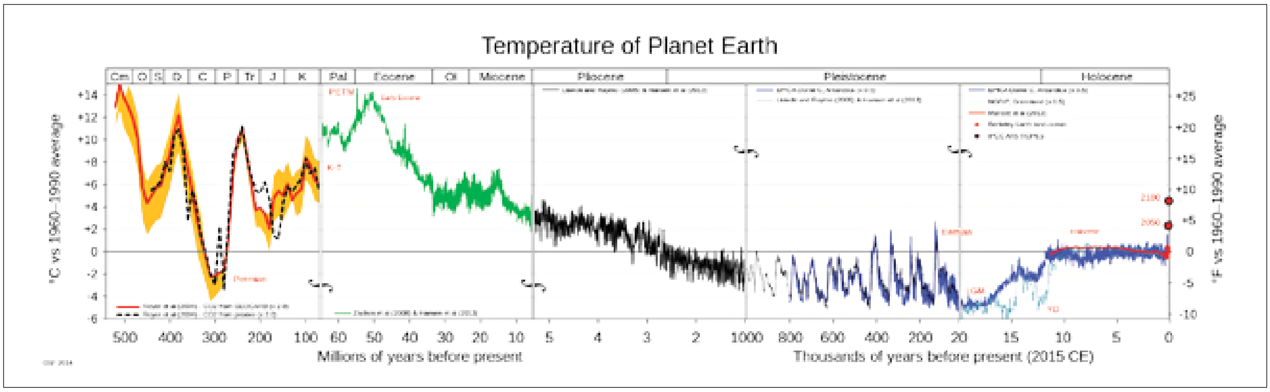


**Total Matter in The Universe**
Researchers have used the most precise measurements yet to derive the proportion of matter. **http://dgit.in/oct20-33**



**United Arab Emirates spacecraft**
Sheikh Mohammed bin Rashid Al Maktoum said UAE plans to send a spacecraft to the moon in 2024. **http://dgit.in/oct20-34**

Credit: Glen Fergus




**Temperature of the planet over the past 500 million years**

heat in the atmosphere, giving rise to the "greenhouse" conditions. The concentration of carbon in the atmosphere was 409.8 parts per million ppm, during the CTM, it had spiked to 1000 ppm. The rising sea levels around the world, combined with the increasing volcanism introduced a large amount of nutrients in the oceans, which gave rise to massive phytoplankton blooms. These phytoplankton blooms which could have dwarfed the continents of today, pumped up a large volume of oxygen into the atmosphere, which then allowed frequent wildfires to rage across the planet. When the volcanism reduced, the amount of nutrients in the oceans did too, the phytoplankton blooms died off, sinking to the seafloor, and providing us with one of the most important resources of the industrial world – oil.

About 50 million years ago, was the PETM. The crocodiles and palm trees near the polar regions were found from this time. Fossil remains of floating ferns, today found only in subtropical regions, has also been found near the north pole. The PETM is more mysterious as we do not know exactly what caused an increase of carbon dioxide being pumped into the atmosphere. Volcanic activity, a cometary impact and the combustion of large quantities of peat have all been suggested as possible sources. The temperature difference between the poles and the tropics was not as pronounced as it is now, which is something of a paradox that puzzles scientists. For heat to move across the planet in large enough scales to maintain a more homogeneous temperature, the temperature differences have to be pronounced. Researchers believe the uniform temperature of the planet had to do with an increasing saturation of water vapour in the atmosphere, which helped transfer the heat across the planet. The steep rise in temperatures around the PETM had another direct influence on humans, it allowed mammalian species, including horses, cows and primates to spread across the planet and occupy more niches.

## ICEHOUSE

Periodically, the Earth converts into what can essentially be called a snowball. The polar ice sheets extend throughout the planet, extending into growing glaciers that are slow moving rivers of ice. These massive, cold tentacles spread across the planet, covering everything in their path. Just as the hothouse phase is accompanied by high sea levels and water vapour in the atmosphere, the icehouse phase is accompanied by low sea levels, with most of the water trapped in the glacial ice sheets. It is even possible that the surface of the oceans were frozen. Instead of forest fires, massive snowstorms and paradoxically dry cyclones raged across the planet. While there were many periods in the history of the planet when it was much colder than it is today, two periods stand out, the Last Glacial Maximum (LGM), commonly known as the Ice Age, and the Late Paleozoic icehouse, also known as the Karoo ice age. All it takes for the planet to go into an icehouse state, is for it to be around six degrees Celsius colder in average temperature. That was how cold it was during the LGM. This should give an idea of how delicate the temperature balance is. Rain, when it fell at all, was also frozen. The atmosphere was extremely dry, with some regions getting only 10 percent of the rainfall they see today. During this time, the vegetation receded across the planet, while the deserts expanded.

The Karoo ice age was similar, but was caused because of a very interesting chain of events. A new category of oxygen breathing land plants and trees began pumping vast amounts of oxygen into the atmosphere. This reduced the proportion of carbon dioxide, which made the planet cooler, causing the polar ice sheets to spread. The ice bounced off even more incident heat energy, creating a positive feedback loop for the ice. The planet stayed in an icehouse state till the termites evolved to take advantage of all the wood. Small methanogenic bacteria piggybacking in the stomachs of the termites, then provided the planet with an essential greenhouse gas needed to get out of the icehouse state – methane.

The temperature history of the Earth shows that it has always been life forms that were responsible for the buildup of greenhouse gases in the atmosphere. The fossil fuels that we burn today are remains of these very life forms.